

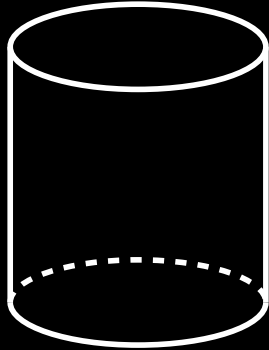
Calculating the Volume of Cylinders

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A **cylinder** is a solid figure with two congruent circular bases connected by a curved surface.

Parts of a Cylinder

circular base



curved surface

circular base

Volume

The **volume** of a cylinder is the amount of space inside it.

Volume is measured in **cubic units**, such as cm^3 , m^3 , or in^3 .

Base

The **base** of a cylinder is a circle with area

$$A = \pi r^2$$

Height

The **height** is the perpendicular distance between the two circular bases.

Measure it straight from one base to the other, at a right angle (90°).

Volume Formula

The formula for the **volume of a cylinder** is

$$V = \pi r^2 h$$

where r is the radius and h is the height.

$$V = \text{Base Area} \times \text{Height}$$

Diameter Given?

If the **diameter** is given, first find the radius using

$$r = \frac{d}{2}$$

For example, if $d = 10$ cm, then $r = 5$ cm.

Value of π

Use $\pi \approx 3.14$ unless another value of π is specified.

Another common value is

$$\pi \approx \frac{22}{7}$$

Use the Same Units

The radius and the height must be in the **same unit**.

Convert first, then compute.

$$1 \text{ m} = 100 \text{ cm}$$

Steps to Find the Volume

Find r (use $r = \frac{d}{2}$ if needed).

Identify h and use the same units.

Substitute into $V = \pi r^2 h$.

Compute and write the **cubic units**.

Examples

Example 1

Find the volume of a cylinder with radius 3 cm and height 7 cm. Use $\pi \approx 3.14$.

$$\begin{aligned} V &= \pi r^2 h \\ &\approx 3.14(3)^2(7) \\ &\approx 3.14(63) \\ &\approx 197.82 \text{ cm}^3 \end{aligned}$$

Example 2

Find the volume of a cylinder with diameter 10 cm and height 6 cm. Use $\pi \approx 3.14$.

$$r = \frac{10}{2} = 5 \text{ cm}$$

$$\begin{aligned} V &= \pi r^2 h \\ &\approx 3.14(5)^2(6) \\ &\approx 3.14(150) \\ &\approx 471 \text{ cm}^3 \end{aligned}$$

Example 3

Find the volume of a cylinder with radius 14 cm and height 5 cm. Use $\pi \approx \frac{22}{7}$.

$$\begin{aligned} V &= \pi r^2 h \\ &\approx \frac{22}{7}(14)^2(5) \\ &\approx \frac{22}{7}(196)(5) \\ &\approx 3\,080 \text{ cm}^3 \end{aligned}$$

Example 4

A water tank is shaped like a cylinder with radius 2 m and height 3 m. Find its volume. Use $\pi \approx 3.14$.

$$\begin{aligned} V &= \pi r^2 h \\ &\approx 3.14(2)^2(3) \\ &\approx 3.14(12) \\ &\approx 37.68 \text{ m}^3 \end{aligned}$$

Example 5

A pipe has an inside radius of 5 cm and a height of 1 m. Find its volume. Use $\pi \approx 3.14$.

$$1 \text{ m} = 100 \text{ cm}$$

$$\begin{aligned} V &= \pi r^2 h \\ &\approx 3.14(5)^2(100) \\ &\approx 7850 \text{ cm}^3 \end{aligned}$$

Example 6

A cylinder has a volume of $1\,570\text{ cm}^3$ and a radius of 5 cm . Find its height. Use $\pi \approx 3.14$.

$$V = \pi r^2 h$$

$$1\,570 \approx 3.14(5)^2 h$$

$$1\,570 \approx 78.5h$$

$$h \approx 20\text{ cm}$$

Practice Exercises

Practice 1

Find the volume of the cylinder.

A cylinder with radius 4 cm and height 10 cm.

Practice 2

Find the volume of the cylinder.

A cylinder with radius 6 cm and height 8 cm.

Practice 3

Find the volume of the cylinder.

A cylinder with radius 7 m and height 5 m.

Practice 4

Find the volume of the cylinder.

A cylinder with diameter 14 cm and height 15 cm. Use $\pi = \frac{22}{7}$.

Practice 5

Find the volume of the cylinder.

A cylinder with diameter 18 cm and height 9 cm.

Practice 6

Find the volume of the cylinder.

A water tank is shaped like a cylinder with radius 2.5 m and height 4 m.

Practice 7

Find the volume of the cylinder.

A candle is cylindrical with radius 3 cm and height 18 cm.

Practice 8

Find the volume of the cylinder.

A cylindrical can has diameter 12 cm and height 20 cm.

Practice 9

Find the volume of the cylinder.

A pipe has an inside radius of 15 cm and height 2 m.

Practice 10

A cylindrical container has a volume of $1\,256\text{ cm}^3$ and a radius of 10 cm . Find its height.

Activities

Activity 1

Find the volume of the cylinder.

A cylinder with radius 5 cm and height 9 cm.

Activity 2

Find the volume of the cylinder.

A cylinder with radius 8 cm and height 7 cm.

Activity 3

Find the volume of the cylinder.

A cylinder with radius 9 m and height 4 m.

Activity 4

Find the volume of the cylinder.

A cylinder with diameter 16 cm and height 12 cm. Use $\pi = \frac{22}{7}$.

Activity 5

Find the volume of the cylinder.

A cylinder with diameter 20 cm and height 14 cm.

Activity 6

Find the volume of the cylinder.

A cylindrical storage drum has radius 3 m and height 5 m.

Activity 7

Find the volume of the cylinder.

A roll of paper is cylindrical with radius 4 cm and height 15 cm.

Activity 8

Find the volume of the cylinder.

A cylindrical bucket has diameter 18 cm and height 16 cm.

Activity 9

Find the volume of the cylinder.

A cylindrical pillar has radius 20 cm and height 3 m.

Activity 10

A cylindrical container has a volume of 2512 cm^3 and a radius of 10 cm . Find its height.

Thank you!